AI for All or AI for Ready Schools?

Implementation Capacity, Early-Warning Support Need, and Public Monitoring in Hong Kong Secondary Schools

Data Science of Politics and Public Administration project

May 2026

Executive Summary

This report examines whether Hong Kong secondary schools show uneven publicly observable capacity to absorb and implement AI education policy. The analysis is framed as a politics and public administration study: the central concern is not whether schools are good or bad, but whether equal funding can become equal student opportunity when schools differ in planning capacity, digital infrastructure, teacher development, governance, and public reporting practices.

The report uses a support-prioritisation framework rather than a ranking framework. The Foundational AI Implementation Capacity Index captures visible conditions that may help schools absorb future AI funding. The Early AI-Specific Signal Index captures direct AI-related public evidence, but it is interpreted cautiously because the funding programme had not yet rolled out at school level in the collected evidence. The Early-Warning Support Need Index identifies candidates for validation and support rather than schools to blame.

The evidence base covers 441 secondary schools. The crawler collected public website evidence for 440 school websites, including 8,231 HTML pages and 1,931 PDFs. After text extraction, 8,763 documents contained usable text. The final analytical dataset retained 12,997 cleaned evidence items after deduplication and confidence weighting.

Headline results

Mean foundational capacity is 27.44/100 and mean early AI-specific signal is 13.77/100. The median early AI-specific signal is 8.33/100, meaning direct AI evidence is still thin for many schools. Mean early-warning support need is 76.86/100, while publication visibility is strongly associated with index values. These results support tiered support and manual validation, not public ranking.

The main policy implication is that AI for All requires Capacity for All. A one-off school grant can help, but it will not automatically produce equal implementation. Public monitoring should therefore combine a small standardised reporting system, support tiers, teacher development, governance guidance, infrastructure support, and manual validation of low-visibility cases.

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1. Introduction

Artificial intelligence is becoming a major education policy priority in Hong Kong. The policy agenda is not only about buying software. It requires school planning, teacher confidence, digital infrastructure, curriculum integration, student learning opportunities, procurement decisions, and responsible use safeguards. That makes AI education a public administration problem: policy outcomes depend on whether schools can convert central support into local implementation.

Hong Kong's AI education agenda is moving quickly. The 2025 Policy Address announced HK$2 billion under the Quality Education Fund to support digital education, including AI literacy, AI education in the core curriculum, and teacher training. EDB Circular Memorandum No. 221/2025 then introduced the AI for Empowering Learning and Teaching Funding Programme, under which eligible publicly funded schools may receive a one-off HK$500,000 grant.

The research asks a practical question: which schools may need more validation and support before funding can be converted into meaningful AI-assisted teaching and student opportunity? The analysis does not claim to measure actual classroom practice. It measures publicly observable evidence and treats low evidence as a candidate signal for follow-up, especially when publication visibility is also low.

- How much foundational capacity is visible in public evidence?
- How much direct AI-specific activity is visible before policy rollout?
- Which dimensions of implementation capacity appear weakest?
- Do support needs appear uneven across districts and school characteristics?
- How should policymakers use the evidence without creating school rankings?

2. Policy and Literature Framework

2.1 Policy implementation capacity

Policy implementation theory emphasises that policy outputs do not automatically become policy outcomes. Schools are frontline organisations. They translate policy into plans, training, procurement, curriculum resources, classroom practice, and monitoring routines. This report therefore treats the AI funding programme as an organisational implementation challenge rather than a simple technology adoption problem.

2.2 Digital inequality and capability

Digital inequality is not only about whether a school has devices. Meaningful use depends on infrastructure, human capacity, social support, institutional routines, and the ability to convert resources into valued opportunities. The capability approach is useful here: equal inputs do not guarantee equal capabilities. A school with strong planning and teacher development may turn HK$500,000 into AI-assisted teaching more quickly than a school that first needs basic infrastructure, procurement guidance, and responsible-use rules.

2.3 Organisational and pedagogical frameworks

The analytical framework draws on SELFIE/DigCompOrg, CFIR, TOE, UNESCO AI competency frameworks, TPACK, digital-divide theory, and capability theory to justify a broader implementation-capacity index. Broad digital indicators such as STEM, e-learning, coding, robotics, and digital platforms are retained, but they are treated as foundational capacity rather than direct AI implementation. This prevents the index from overclaiming that general digital activity is already AI practice.

3. Data and Methodology

Table 1. Public evidence pipeline

Step	Count	Interpretation
Secondary schools in baseline	441	Official school-level analytical frame
School websites successfully collected	440	Public school website evidence
HTML pages saved	8,231	Website pages and public web records
PDFs saved	1,931	Public PDF documents
Usable documents after text extraction	8,763	Documents with extractable text
Cleaned evidence items	12,997	Deduplicated public evidence snippets
Schools with at least one cleaned evidence item	341	Not a performance count; partly affected by publication visibility

Note. Counts are taken from the processed data files. OCR was not used, so scanned PDFs may be underrepresented.

The evidence cleaning process removes near-duplicate evidence by school, source URL, inferred dimension, inferred sub-dimension, and snippet signature. Evidence is weighted by confidence: high-confidence evidence receives greater weight than medium- or low-confidence evidence. Each school-dimension score is capped on a small ordinal scale before conversion to 0-100, so a long archive cannot dominate a dimension simply by repeating the same keyword many times.

This approach is appropriate for early monitoring, but it is not a substitute for direct school reporting. Website practices differ: some schools publish detailed reports, archived PDFs, and long web pages; others may be active in practice but publish less. For this reason, the analysis includes a publication visibility layer and uses cautious language such as public evidence, visible signal, candidate for follow-up validation, and potential support need.

4. Index Construction

The index structure follows an early-warning support logic rather than a school-ranking logic. It distinguishes between capacity to absorb future funding, early direct AI signals, quality of AI use, support need, and publication visibility.

Table 2. Index components

Index layer	What it measures	How to interpret it
Foundational AI Implementation Capacity	Visible enabling conditions such as planning, digital infrastructure, teacher development, curriculum integration, student opportunities, governance, and resource mobilisation.	Capacity to absorb future AI funding; not direct proof of AI implementation.
Early AI-Specific Signal	Direct AI-related public evidence such as AI literacy, AI-assisted teaching, AI tools, AI teacher development, AI assessment, responsible AI, cross-subject AI use, and procurement.	Early signal only. Low evidence before rollout is not school failure.
AI Quality of Use	A diagnostic layer focusing on deeper use: teacher development, cross-subject use, responsible AI, AI literacy, assessment, and AI-assisted teaching.	Quality-oriented public signal, not full lesson-level evaluation.
Early-Warning Support Need	A support triage score combining low foundational capacity, teacher-development gap, governance gap, infrastructure gap, and low early AI-specific signal.	Candidate need for validation and support; not a school grade or ranking.
Evidence Confidence / Publication Visibility	Document volume, usable text, evidence source count, high-confidence evidence, and dimensions with evidence.	Reliability and publication-practice diagnostic; not school quality.

Table 3. Conceptual dimension map for public evidence coding

Dimension	Why it belongs	Framework support	Public evidence examples
Strategic leadership and planning	AI funding requires planning, not just buying tools.	DigCompOrg; TOE; capability theory	School plan; annual plan; major concerns; AI or digital strategy; development plan.
Foundational digital infrastructure	Schools need basic digital capacity before AI can work.	SELFIE/DigCompOrg; TOE; digital divide theory	E-learning platforms; LMS; Google Classroom; Microsoft Teams; BYOD; tablets; computer rooms; smart classrooms; AI lab.
Teacher development and human capacity	AI implementation depends heavily on teachers' confidence and training.	SELFIE; TPACK; capability theory	Teacher training; professional development; staff workshop; AI teacher training; communities of practice.
Curriculum and pedagogical integration	AI should enter teaching and learning, not only administration or procurement.	TPACK; DigCompOrg teaching and learning practices	STEM; STEAM; coding; robotics; AI literacy; cross-subject learning; project-based learning; digital pedagogy.
Student digital and AI learning opportunities	The policy aim is student learning opportunity, not only school capacity.	Capability theory; digital divide theory	AI literacy activities; competitions; innovation activities; coding teams; robotics teams; digital ambassadors.
Governance, responsible use, and safeguards	AI creates risks around privacy, ethics, academic integrity, and data use.	DigCompOrg; capability theory; responsible AI governance	Privacy; cybersecurity; data security; digital citizenship; responsible AI; AI ethics; academic integrity; acceptable-use policy.
Implementation administration and resource mobilisation	Some schools may have ideas but lack procurement or project-management capacity.	TOE; public administration implementation theory	Procurement; tenders; subscriptions; grant use; vendor or platform selection; implementation reports.
Equity and contextual conversion risk	Equal funding may not become equal capability across schools.	Digital divide theory; capability theory	District income; public rental housing share; fee/remission evidence; NCS support; SEN support; finance type; resource constraints.

Note. The first seven dimensions feed the main school-level public-evidence index. District socioeconomic context is used as a sensitivity layer because it is district-level, not school-level.

The main Early-Warning Support Need formula uses five observable school-evidence components: 25% low foundational capacity, 20% teacher development gap, 15% governance/responsible-use gap, 15% infrastructure gap, and 10% low early AI-specific signal. These components add to 85% of the full conceptual model and are rescaled to 100% in the primary no-context version. A separate context-sensitivity version includes a 15% district contextual need component.

Important interpretation rule
The support-need score is not a grade. It is a triage signal. A school may have low visible public evidence because it publishes little online, because activities are not yet underway, or because it needs support. The appropriate administrative response is validation and support, not labelling.

5. Findings

5.1 Headline distributions: capacity and direct AI signals remain low and uneven

Table 4. Overall distribution of monitoring indices

Metric	Mean	Median	P25	P75	P90	Gini
Foundational AI Implementation Capacity	27.44	19.05	4.76	42.86	76.19	0.566
Early AI-Specific Signal	13.77	8.33	0.00	20.83	37.50	0.627
AI Quality of Use	16.93	5.56	0.00	27.78	44.44	0.634
Early-Warning Support Need	76.86	87.96	64.85	98.60	100.00	0.183

Metric	Mean	Median	P25	P75	P90	Gini
Evidence Confidence / Publication Visibility	50.11	53.31	25.98	72.94	85.30	0.303

Note. All index values are on a 0-100 scale. Higher support need means stronger candidate need for validation and support, not lower school quality.

Mean foundational capacity is 27.44/100, and the median is 19.05/100. Mean early AI-specific signal is only 13.77/100, with a median of 8.33/100. Quality of use is similarly thin, with a mean of 16.93/100 and median of 5.56/100. The Gini values for early AI signal (0.627) and quality (0.634) indicate substantial unevenness in visible direct AI evidence.

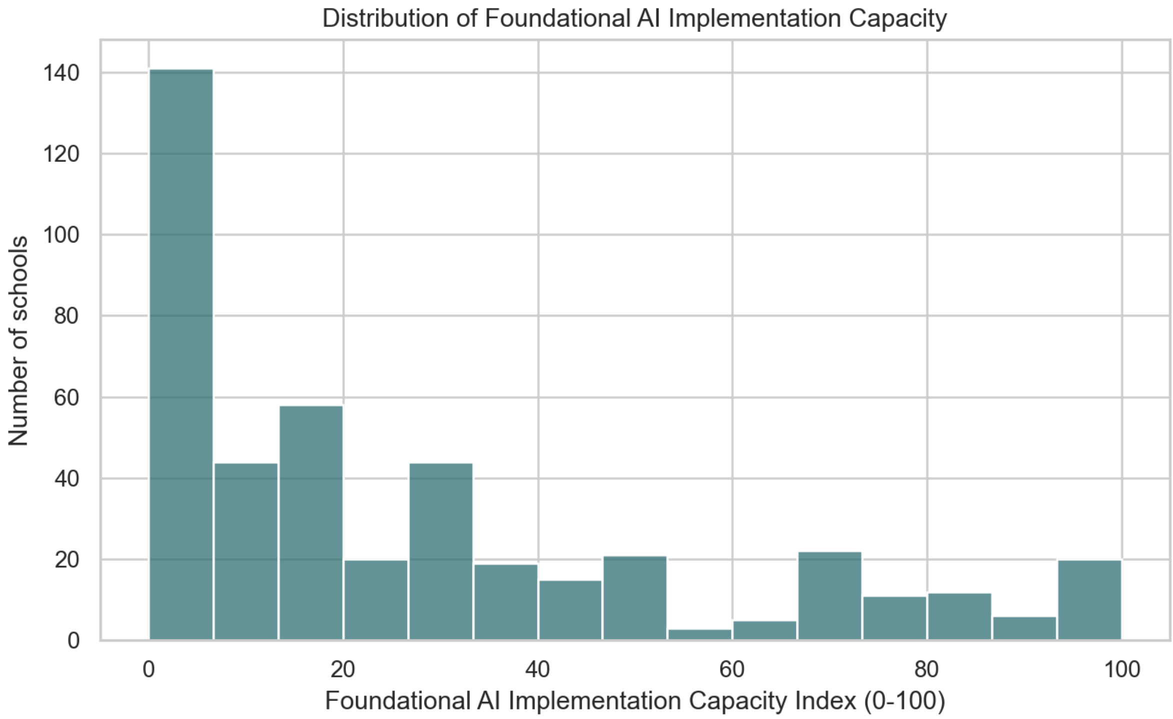

Figure 1. Distribution of Foundational AI Implementation Capacity scores.

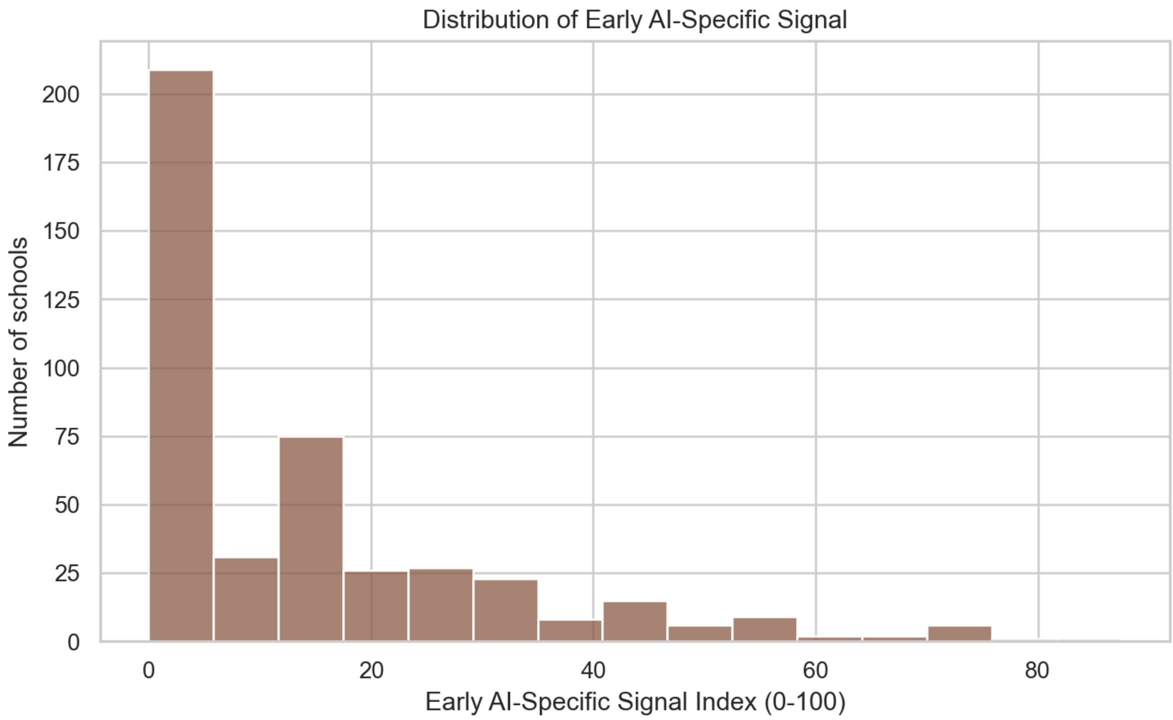

Figure 2. Distribution of Early AI-Specific Signal scores.

5.2 Foundational capacity is strongest in curriculum signals, but weaker in planning, infrastructure, governance, and teacher development

Table 5. Foundational capacity dimensions

Dimension	Schools with evidence (%)	Mean score	Evidence items
Curriculum and pedagogical integration	60.09	43.46	3,055
Student digital / AI learning opportunities	55.78	41.65	1,954

Dimension	Schools with evidence (%)	Mean score	Evidence items
Teacher development and human capacity	36.73	26.61	1,869
Foundational digital infrastructure	36.05	23.81	1,333
Administrative / resource mobilisation capacity	35.15	24.79	1,155
Governance and responsible use	29.93	16.93	406
Strategic leadership and planning	24.04	14.81	675

Curriculum and pedagogical integration is the most visible foundational dimension: 60.09% of schools have evidence, with a mean score of 43.46. Student digital or AI learning opportunities are also relatively visible at 55.78%. In contrast, strategic leadership and planning appears in only 24.04% of schools, governance and responsible use in 29.93%, foundational digital infrastructure in 36.05%, and teacher development capacity in 36.73%.

This pattern matters because the policy does not only require enthusiasm for AI. It requires schools to plan, train teachers, procure or manage tools, establish responsible-use routines, and integrate AI into teaching and learning. The public evidence suggests that the governance-heavy and implementation-administration parts of capacity are less visible than broad curriculum or innovation signals.

5.3 Direct AI-specific evidence is still thin

Table 6. Early AI-specific signal dimensions

Dimension	Schools with evidence (%)	Mean score	Evidence items
Cross-subject AI use	52.61	37.72	1,740
AI literacy activities	31.75	14.97	428
AI-assisted teaching	26.08	15.95	669
AI assessment	21.54	12.55	604
AI teacher development	21.09	13.76	497
AI procurement	13.15	7.33	233
Responsible AI	11.11	6.65	138
AI tools / platforms	3.40	1.21	21

Cross-subject AI use is the most visible direct AI dimension, with evidence in 52.61% of schools and a mean score of 37.72. AI literacy activities appear in 31.75% of schools. However, direct evidence of AI tools or platforms appears in only 3.40%, responsible AI in 11.11%, AI procurement in 13.15%, AI teacher development in 21.09%, and AI assessment in 21.54%.

This does not mean schools have failed to implement AI. The policy period is early, and public evidence is incomplete. The appropriate interpretation is that direct AI-specific signals are still thin and uneven, so schools may need differentiated support before being expected to convert grants into multi-subject AI-assisted teaching.

5.4 Percentile-based support tiers avoid labelling hundreds of schools as high risk

Table 7. Percentile-based support-prioritisation tiers

Tier	Meaning	Schools	Mean support need	Mean capacity	Mean early AI signal	Mean visibility
Tier 1	Tier 1: Immediate validation / intensive support candidate	45	100.00	0.00	0.00	14.60

Tier	Meaning	Schools	Mean support need	Mean capacity	Mean early AI signal	Mean visibility
Tier 2	Tier 2: Targeted capacity-building candidate	66	99.83	0.29	0.69	16.66
Tier 3	Tier 3: General monitoring / standard support	220	86.03	19.96	10.45	51.79
Tier 4	Tier 4: Lower immediate support need / peer-learning candidate	110	35.27	69.91	33.86	81.35

Note. Tier 1 is the top 10% of support need, Tier 2 the next 15%, Tier 3 the middle 50%, and Tier 4 the lowest 25%.

The tiering rule is deliberately percentile-based. This approach identifies differentiated support pathways without turning the index into a broad risk label. Tier 1 contains 45 schools, Tier 2 contains 66, Tier 3 contains 220, and Tier 4 contains 110. Tier 1 and Tier 2 should be interpreted as candidates for validation and support, not as a public list of poor performance.

The capacity-signal scatterplot gives a complementary diagnostic view. Using median thresholds, 193 schools show higher foundational capacity and higher early AI-specific signal; 28 show higher capacity but lower early AI-specific signal; 39 show lower foundational capacity but higher early AI-specific signal; and 181 show lower public evidence on both axes.

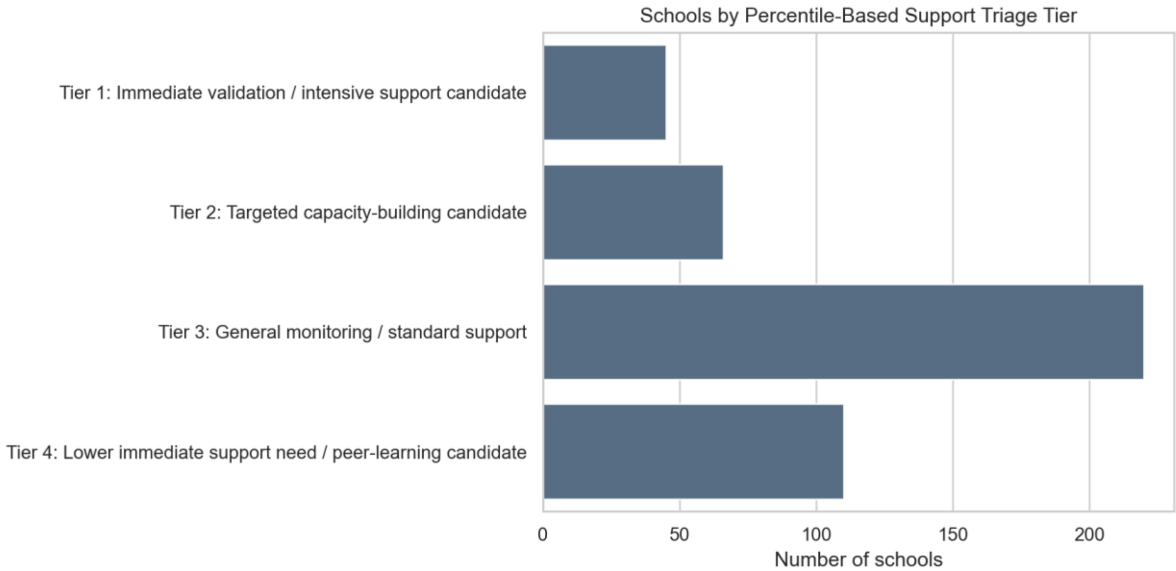

Figure 3. Schools by percentile-based support triage tier.

5.5 District patterns are visible, but district is not the cause

Table 8. District support-need patterns

Category	District	Schools	Mean capacity	Mean early AI	Mean support need	Tier 1/2 share
Highest mean support need	Islands	9	11.64	7.41	89.17	55.6%
Highest mean support need	North	20	14.52	5.00	88.08	40.0%
Highest mean support need	Tai Po	20	21.43	10.21	83.97	15.0%
Highest mean support need	Central & Western	11	22.08	12.50	81.88	18.2%
Highest mean support need	Sham Shui Po	25	24.76	13.17	80.42	28.0%
Lowest mean support need	Kwun Tong	33	33.33	19.32	69.74	18.2%

Category	District	Schools	Mean capacity	Mean early AI	Mean support need	Tier 1/2 share
Lowest mean support need	Eastern	30	33.81	16.67	69.79	30.0%
Lowest mean support need	Wong Tai Sin	22	34.63	18.75	70.05	22.7%
Lowest mean support need	Wan Chai	16	32.44	16.67	72.20	25.0%
Lowest mean support need	Yuen Long	39	31.38	13.89	74.05	20.5%

Note. District comparisons are descriptive. They should be interpreted with school-level validation and publication-visibility checks.

Mean support need is highest in Islands (89.17), North (88.08), Tai Po (83.97), Central & Western (81.88), and Sham Shui Po (80.42). It is lowest in Kwun Tong (69.74), Eastern (69.79), Wong Tai Sin (70.05), Wan Chai (72.20), and Yuen Long (74.05). The highest-lowest district gap in mean support need is 19.43 points. This is a descriptive pattern, not proof that district itself causes implementation capacity.

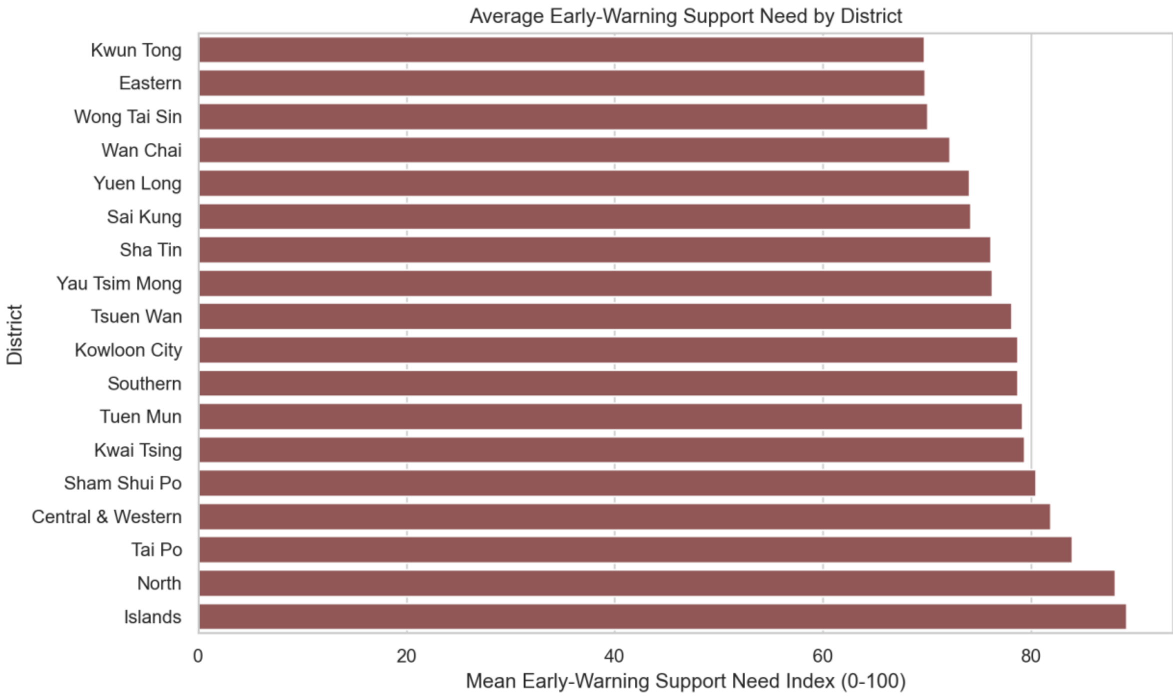

Figure 4. Average Early-Warning Support Need by district.

District socioeconomic context is included only as a sensitivity layer. District-level census variables are useful background information, but they do not describe individual schools. When contextual need is included, district-level socioeconomic indicators correlate more strongly with the context-sensitive support score. For example, median monthly employment earnings correlates negatively with mean context-sensitive support need (Spearman rho = -0.710), while the district contextual need index correlates positively (rho = 0.707).

5.6 Publication visibility is a major trust issue

Table 9. Publication visibility correlations

Publication variable	Outcome	Spearman rho
Publication visibility	Foundational capacity	0.955
Publication visibility	Early AI-specific signal	0.868
Publication visibility	AI quality of use	0.856
Publication visibility	Early-warning support need	-0.927
Total extracted text length	Foundational capacity	0.888

Publication variable	Outcome	Spearman rho
Total extracted text length	Early-warning support need	-0.858
Publication volume index	Foundational capacity	0.781
Publication volume index	Early-warning support need	-0.755

Note. The listed correlations show association, not causality.

Publication visibility is strongly associated with the monitoring scores. The visibility index correlates 0.955 with foundational capacity, 0.868 with early AI-specific signal, 0.856 with quality, and -0.928 with support need. Total extracted text length has similar associations. This means the index partly captures whether schools publish evidence publicly. That is not a defect if the tool is used for triage, but it would be a serious problem if it were used as a public league table.

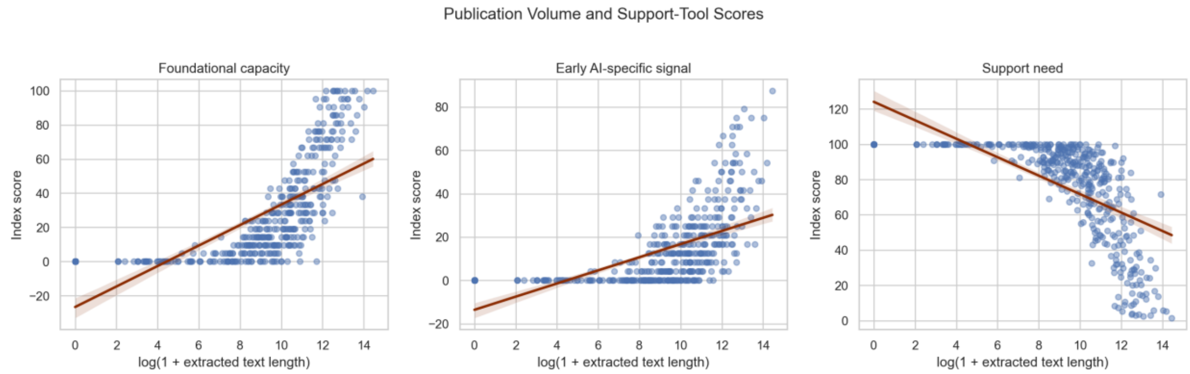

Figure 5. Publication volume and support-tool scores.

Publication-adjusted district summaries change the district ordering. For example, Islands and North have the highest unadjusted support need means, but after publication adjustment Tsuen Wan and Tai Po move higher. This reinforces the report's central caution: public website monitoring is useful, but incomplete.

5.7 School characteristics matter, but they do not replace direct evidence

Table 10. Selected school-characteristic summaries

Characteristic	Group	Schools	Mean capacity	Mean early AI	Mean support need	Tier 1 share
School Type	Aided	354	27.60	13.58	76.64	9.9%
School Type	CAPUT	2	2.38	2.08	99.06	50.0%
School Type	DSS	56	21.51	12.28	82.63	16.1%
School Type	Gov't	29	38.59	19.68	66.86	0.0%
Finance Type	AIDED	340	27.62	13.47	76.60	10.3%
Finance Type	CAPUT	2	2.38	2.08	99.06	50.0%
Finance Type	DIRECT SUBSIDY SCHEME	54	20.55	12.12	83.56	16.7%
Finance Type	GOVERNMENT	27	39.33	20.68	66.39	0.0%
Finance Type	Unknown	18	29.63	15.28	75.00	0.0%

Note. Group comparisons are descriptive and should not be treated as causal explanations.

The analysis identifies statistically detectable differences across school types for early-warning support need and foundational capacity. However, these are group-level patterns. They should not be used to infer individual school need without validation. The analysis also shows that publication visibility and evidence volume are among the strongest correlates of the index outcomes, which is another reason to rely on manual checking before making support decisions.

6. Discussion

6.1 Main interpretation: a support-conversion problem

The evidence supports a clear public administration interpretation. Hong Kong's AI education policy is not only a funding question; it is a support-conversion question. Schools differ in visible capacity to convert money, guidance, and policy expectations into teacher practice and student opportunity. Equal funding can therefore produce unequal implementation unless support is matched to starting capacity.

6.2 Why the report should avoid ranking language

The index should not be described as a ranking, grading tool, or proof of poor performance. A low public signal can mean several things: low implementation, unpublished implementation, sparse website publication, poor extractability of documents, or evidence that requires manual review. The phrase Early-Warning Support Need is therefore more accurate than Equity Risk. It keeps the focus on administrative response: validation, capacity-building, teacher support, governance support, and follow-up.

6.3 Why publication visibility should remain visible

The Evidence Confidence / Publication Visibility layer is essential. It helps prevent website publication practice from being confused with school quality. Schools with richer websites and longer archives are more likely to show public evidence. Schools with lower visibility should not automatically be assumed to have lower actual practice. Instead, low visibility plus high support need should trigger follow-up validation.

7. Policy Recommendations

1. Use the tiers as support pathways, not as public rankings. Tier 1 schools should be candidates for immediate validation and intensive support; Tier 2 for targeted capacity-building; Tier 3 for general monitoring and standard support; and Tier 4 for lower immediate support need or voluntary peer learning.
2. Pair funding with basic implementation scaffolds. Provide planning templates, procurement guidance, sample responsible-use policies, teacher development sequences, and subject-level AI-assisted teaching examples.
3. Create a small standardised reporting requirement for funded schools. Useful indicators include subjects using AI-assisted teaching, teacher training completed, teaching resources developed, student AI literacy activities, procurement choices, and responsible-AI governance arrangements.
4. Validate before acting. Use public evidence to shortlist candidates for follow-up, then verify through school self-report, document review, or sampling. Low publication visibility should increase the need for validation, not blame.
5. Invest in governance and responsible use. The low visibility of governance, responsible AI, procurement, and infrastructure signals suggests that schools need practical guidance on privacy, data security, academic integrity, bias, assessment integrity, age-appropriate use, and teacher oversight.
6. Use peer learning carefully. Tier 4 schools or high-capacity/high-signal schools may be invited into voluntary demonstration, mentoring, and resource-sharing roles. This should not become a prestige ranking or reputational pressure mechanism.

8. Limitations and Validation Needs

- Website evidence is incomplete. Absence of public evidence does not mean absence of activity.
- Publication practices vary strongly across schools. The visibility correlations show that document volume and text length shape observed scores.

- Keyword-derived evidence requires manual validation. Even after deduplication and confidence weighting, snippets can still be weak, duplicated in meaning, or context-dependent.
- District socioeconomic context is district-level. It should not be interpreted as school-level student socioeconomic composition.
- The index predates full rollout of the funding programme. Direct AI-specific evidence should therefore be read as early signal, not implementation failure.

The most important next step is manual validation. A reasonable validation design would sample schools from all four support tiers and from low-visibility/high-support-need cases. Reviewers should check whether evidence snippets genuinely indicate the coded dimension, whether the school website is unusually sparse or unusually rich, and whether future EDB implementation reports confirm or contradict the public evidence.

9. Conclusion

The analysis strengthens the project's public administration contribution. The key finding is not that some schools should be ranked above others. The key finding is that publicly observable implementation capacity is low, uneven, and deeply affected by publication visibility. AI education policy will be more equitable if funding is paired with tiered support, validation, teacher development, governance guidance, and practical reporting.

AI for All therefore requires Capacity for All. The purpose of the index is to identify where implementation support may need to arrive first, so that AI education funding does not quietly become AI for already-ready schools.

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